

Shikher Sharma

Postdoctoral Fellow in Mathematics

Department of Mathematics, Technion – Israel Institute of Technology, Haifa, Israel

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Google Scholar | ResearchGate

Research Profile

My research focuses on optimization and nonlinear analysis on Hadamard manifolds, with particular emphasis on equilibrium problems, variational inequalities and inclusions, fixed point methods, and iterative optimization algorithms. My work also includes applications of optimization techniques to machine learning and signal processing.

Research interests: Optimization on Hadamard manifolds; equilibrium problems; variational inequalities and inclusions; fixed point theory; nonlinear optimization; machine learning; signal processing.

Academic Appointments

Postdoctoral Fellow

Department of Mathematics, Technion – Israel Institute of Technology

April 2025 – Present

Haifa, Israel

Advisor: Prof. Simeon Reich

Education

Banaras Hindu University

Ph.D. in Mathematics

2024

Varanasi, India

Supervisor: Prof. D. R. Sahu

Thesis: *Applications of Fixed Point Analysis on Hadamard Spaces*

Hansraj College, University of Delhi

M.Sc. in Mathematics

2018

New Delhi, India

Deen Dayal Upadhyaya College, University of Delhi

B.Sc. (Honours) in Mathematics

2016

New Delhi, India

Publications

1. **Shikher Sharma** and Simeon Reich, “A Bregman Regularized Proximal Point Method for Solving Equilibrium Problems on Hadamard Manifolds,” *Rendiconti del Circolo Matematico di Palermo*, 75(5), 122, 2026.
doi:10.1007/s12215-026-01446-2
2. D. R. Sahu, **Shikher Sharma**, and Pankaj Gautam, “Efficient Douglas–Rachford Methods on Hadamard Manifolds with Applications to the Heron Problems,” *Applied Mathematics & Optimization*, 93, 78, 2026.

doi:10.1007/s00245-026-10402-6

3. D. R. Sahu and **Shikher Sharma**, “A derivative free projection method for the singularity of vector field with convex constraint on Hadamard manifolds,” *Optimization Methods and Software*, 40(2), 266–286, 2025.
doi:10.1080/10556788.2024.2400700
4. D. R. Sahu, **Shikher Sharma**, J. C. Yao, and Xiaopeng Zhao, “Accelerated iterative splitting methods on Hadamard manifolds,” *Optimization*, 74(16), 4169–4206, 2025.
doi:10.1080/02331934.2024.2399807
5. D. R. Sahu, Ariana Pitea, **Shikher Sharma**, and Amit Kumar Singh, “Applications of a variable anchoring iterative method to equation and inclusion problems on Hadamard manifolds,” *Communications in Nonlinear Science and Numerical Simulation*, 138, 1–27, 2024.
doi:10.1016/j.cnsns.2024.108192
6. D. R. Sahu, Feeroz Babu, and **Shikher Sharma**, “A new self-adaptive iterative method for variational inclusion problems on Hadamard manifolds with applications,” *Numerical Algorithms*, 94, 1435–1460, 2023.
doi:10.1007/s11075-023-01542-9
7. D. R. Sahu, Feeroz Babu, and **Shikher Sharma**, “The S-iterative techniques on Hadamard manifolds and applications,” *Journal of Applied and Numerical Optimization*, 2, 353–371, 2020.
doi:10.23952/jano.2.2020.3.06

Preprints and Submitted Work

1. **Shikher Sharma**, Pankaj Gautam, and Simeon Reich, “Regularized Extragradient Methods for Solving Equilibrium Problems on Hadamard Manifolds,” submitted.
arXiv:2506.22391
2. D. R. Sahu, **Shikher Sharma**, Pankaj Gautam, and Simeon Reich, “A Modified Dai–Liao Spectral Conjugate Gradient Method with an Application to Signal Processing,” submitted.
arXiv:2501.15300
3. D. R. Sahu, Amit Kumar Singh, and **Shikher Sharma**, “Normal S-Iterative Approach for Broadcast and Incremental Optimization Algorithms,” submitted.
4. D. R. Sahu, Amit Kumar Singh, and **Shikher Sharma**, “Fixed Point Optimization Algorithm for a Sequence of Mappings with Application to Network Bandwidth Allocation Problems,” submitted.

Invited Talks

- Invited lecture, *Nonlinear Analysis and Optimization Seminar*, Technion – Israel Institute of Technology, Haifa, Israel, July 2026.
- Invited lecture, *Nonlinear Analysis and Optimization Seminar*, Technion – Israel Institute of Technology, Haifa, Israel, February 2026.
- Invited lecture, *Nonlinear Analysis and Optimization Seminar*, Technion – Israel Institute of Technology, Haifa, Israel, September 2025.

- Invited lecture, *Nonlinear Analysis and Optimization Seminar*, Technion – Israel Institute of Technology, Haifa, Israel, May 2025.
- Invited talk, *On-Site Workshop on Control and Optimization II*, Technion – Israel Institute of Technology, Haifa, Israel, May 2025.

Conferences and Workshops

- Annual Meeting of the Israel Mathematical Union, Bar-Ilan University, Israel, July 5–6, 2026.
- 2-Day Workshop on Partial Differential Equations: Analysis and Its Applications to Engineering Problems in Industry, ABV-IIITM, Gwalior, India, November 2024.
- 4th International Conference on Nonlinear Applied Analysis and Optimization (ICNNAO-2024), NIT Hamirpur, India, October 2024.
- 3rd International Conference on Nonlinear Applied Analysis and Optimization (ICNNAO-2023), NIT Durgapur, India, December 2023.
- International Conference on Variational Analysis and Optimization with Applications (ICVAOA-2023), AMU, Aligarh, India, September 2023.
- International Symposium on Applied Optimization and Game-Theoretic Models for Decision Making, ISI Delhi Centre, India, February 2023.
- 2nd International Conference on Nonlinear Applied Analysis and Optimization (ICNNAO-2022), IIT (BHU), Varanasi, India, December 2022.
- National Conference of the Academy for Progress of Mathematics on Recent Advances in Mathematical Analysis and Applications, BHU, Varanasi, India, May 2022.
- International Conference on Nonlinear Applied Analysis and Optimization (ICNNAO-2021), IIT (BHU), Varanasi, India, December 2021.
- International Conference on Recent Trends in Analysis and Optimization (IRCTAO-2020), NITTTR, Bhopal, India, November 2020.

Awards, Fellowships, and Scholarships

- **Simons Foundation Travel Grant for ICM 2026**, 2026.
Travel support awarded by the Simons Foundation in collaboration with the International Mathematical Union (IMU) to attend the International Congress of Mathematicians, Philadelphia, USA.
- **Joint CSIR–UGC NET–JRF Fellowship**, 2019.
All India Rank: 35.
- **Joint CSIR–UGC NET**, 2018.
- **Graduate Aptitude Test in Engineering (GATE)**, 2018.
- **INSPIRE Scholarship**, Department of Science and Technology, Government of India, 2013–2018.

Professional Memberships

- Working Group on Generalized Convexity (WGGC)
- Bilevel Optimization Society (BOS)

Computational Skills

Scientific computing: MATLAB, Python, Mathematica

Typesetting: LaTeX

Languages

Hindi (native); English (fluent)

References

Prof. D. R. Sahu

Department of Mathematics
Banaras Hindu University
Varanasi, India
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Prof. J. C. Yao

Center for General Education

China Medical University
Taichung, Taiwan
yaojc@mail.cmu.edu.tw

Prof. Simeon Reich

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Technion – Israel Institute of Technology
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Dr. Pankaj Gautam

Department of Applied Mathematics and
Scientific Computing
Indian Institute of Technology Roorkee
Roorkee, India
pankaj.gautam@amsc.iitr.ac.in